ARTIFICIAL INTELLIGENCE

Implementation of the ID3 Algorithm

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MSC. COMPUTER SCIENCE

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DECLARATION

By submitting this assignment, I certify that this work is solely my own. All code is original or properly attributed. I have not shared my solution with any other student. I understand the consequences of academic dishonesty as outlined in the University’s student conduct policy.

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Date: 12 June, 2026.

1.0 ABSTRACT

This report presents the design, implementation, and evaluation of a from-scratch ID3 decision tree classifier in Python. The project implements entropy, information gain, recursive tree induction, prediction, cross-validation, discretization, visualization, comparison with scikit-learn, and reduced error pruning. Experiments use the Tennis and Sunburn categorical datasets, with optional Iris analysis after equal-frequency discretization of numerical attributes.

The implementation follows the classical ID3 criterion of selecting the attribute with maximum information gain and represents learned trees as nested dictionaries. Results confirm correct entropy and gain calculations, perfect training accuracy on the small categorical examples, and moderate cross-validation performance reflecting limited sample size.

A pruning audit showed that naive validation-tie pruning can over-collapse trees; a conservative rule was added. The project demonstrates how information theory supports interpretable classification while exposing limitations involving small datasets, unseen values, discretization, and validation-set reliability in decision tree learning workflows.

2.0 INTRODUCTION AND THEORETICAL BACKGROUND

Decision trees classify examples by recursively testing feature values. ID3 selects the attribute that gives the largest information gain, which is the reduction in entropy after a split. Entropy is computed as $H(S) = -\text{sum } p(c) \log_2 p(c)$, so pure nodes have zero uncertainty and mixed nodes have higher uncertainty.

3.0 IMPLEMENTATION DETAILS WITH PSEUDOCODE

The custom implementation uses nested dictionaries for trees and separates loading, model construction, evaluation,

visualization, and pruning into modules.

```
ID3(data, attributes, target, default):
    if data is empty: return default
    if labels are pure: return that label
    if attributes are exhausted: return majority class
    choose attribute with highest information gain
    recurse for each sorted branch value
    return nested dictionary tree
```

4.0 EXPERIMENTAL RESULTS

The performance of the custom ID3 decision tree classifier was evaluated using the Tennis and Sunburn datasets. The experiments examined dataset characteristics, tree complexity, training accuracy, cross-validation performance, and the effect of pruning on the learned model.

4.1 Dataset and Tree Characteristics

Table 4.1 summarizes the datasets used in the experiments and the size of the decision trees generated by the ID3 algorithm.

Dataset	**Number of Records**	**Target Attribute**	**Tree Size (Nodes)**
Tennis	14	Play	8
Sunburn	8	Burnt	7
Iris	150	target_name	Optional

The Iris dataset was included in the implementation but required additional dependencies to be available during execution.

The Tennis dataset generated a decision tree containing eight nodes, while the Sunburn dataset produced a slightly smaller tree consisting of seven nodes. These results indicate that the algorithm was able to construct relatively compact and interpretable classification models from both datasets.

4.2 Classification Performance

To evaluate the predictive capability of the classifier, training accuracy and cross-validation performance were measured.

Performance Metric	Tennis	Sunburn
Training Accuracy	1.0000	1.0000
Cross-Validation Mean Accuracy	0.6667	0.6250
Cross-Validation Standard Deviation	0.2108	0.4146

The results show that the custom ID3 implementation achieved perfect training accuracy on both datasets. This confirms that the generated decision trees correctly classified all training examples. However, cross-validation results were lower, with average accuracies of 66.67% and 62.50% for the Tennis and Sunburn datasets, respectively. This difference suggests that the datasets are too small to provide highly stable estimates of generalization performance. The relatively large standard deviations further indicate variability across folds.

4.3 Pruning Evaluation

A reduced-error pruning experiment was conducted to determine whether simplifying the decision tree could improve model performance.

Metric	Unpruned Tree	Pruned Tree
Training Accuracy	1.0	1.0
Validation Accuracy	1.0	1.0
Test Accuracy	1.0	1.0
Number of Nodes	8	8

The pruning results show that no branches were removed from the final Tennis decision tree. Training, validation, and test accuracies remained unchanged after pruning, and the number of nodes stayed at eight. This indicates that the pruning algorithm correctly determined that additional simplification would not improve predictive performance and therefore preserved the original structure of the model.

4.4 Comparison with scikit-learn

A comparison framework was developed to benchmark the custom ID3 implementation against the scikit-learn Decision Tree classifier.

Dataset	Algorithm	Accuracy	Tree Size	Training Time (ms)
Not Available	Not Available	Not Available	Not Available	Not Available

The comparison module was successfully implemented; however, scikit-learn was unavailable in the execution environment used to generate the project outputs. Consequently, comparison statistics could not be produced at the time of experimentation. Once the required dependencies are installed, this section can be populated automatically using the generated comparison framework.

4.5 Summary of Findings

The experimental results demonstrate that the custom ID3 classifier successfully implemented the core principles of decision tree learning. The model achieved perfect training accuracy on the available categorical datasets and produced compact, interpretable trees. Cross-validation results highlighted the limitations of small datasets, while the pruning experiment confirmed that the conservative pruning strategy avoided unnecessary simplification. Overall, the findings validate the correctness of the implementation and illustrate the practical application of entropy and information gain in decision tree classification.

Iris tree

```
petal width (cm)
├── high
│   └── sepal width (cm)
│       ├── high -> virginica
│       ├── low -> virginica
│       └── medium
│           └── petal length (cm)
│               ├── high
```

```
│           │   └── sepal length (cm)
│           │       ├── high -> virginica
│           │       └── medium -> virginica
│           └── medium
│               └── sepal length (cm)
│                   └── medium -> virginica
├── low -> setosa
└── medium
    └── petal length (cm)
        ├── high
        │   └── sepal length (cm)
        │       ├── high -> virginica
        │       └── medium
        │           └── sepal width (cm)
        │               └── low -> virginica
        └── medium -> versicolor
```

Sunburn tree

```
Height
├── Average -> Sunburnt
├── Medium -> None
├── Short
│   └── Lotion
│       ├── No -> Sunburnt
│       └── Yes -> None
└── Tall -> None
```

Tennis tree

```
outlook
├── Overcast -> 1
├── Rain
|   └── wind
|       ├── Strong -> 0
|       └── Weak -> 1
└── Sunny
    └── humidity
        ├── High -> 0
        └── Normal -> 1
```

5.0 DISCUSSION AND ANALYSIS

Question 1.1 concerns information theory basics. Entropy measures class uncertainty, and information gain measures how much uncertainty is reduced by splitting on an attribute. The implementation uses base-2 logarithms and verifies the expected Tennis Outlook gain of about 0.2467.

Question 5.1 concerns model behavior and evaluation. Perfect training accuracy on Tennis and Sunburn is expected because the datasets are small and categorical, but cross-validation accuracy is lower. The pruning audit showed that tiny validation sets can make pruning too aggressive, so pruning is conservative on validation ties.

6. CONCLUSION AND FUTURE WORK

The project implements the full custom ID3 workflow and demonstrates interpretable classification through entropy, information gain, recursive tree construction, evaluation, visualization, and pruning. Future work should add stratified validation, direct continuous threshold selection, larger datasets, richer metrics, and more automated tests.

7.0 REFERENCES

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